

Latent to Language

Latent to Language: Psychometric Context Engineering for LLM-Based Imputation

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The Problem

When respondents skip items within a psychometric scale, the latent score cannot be computed without imputation.

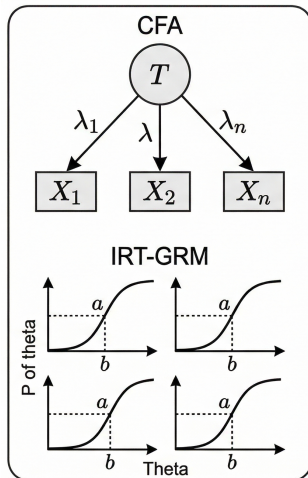
Traditional methods impute from observed correlations, but they never see the measurement model that defines the construct.

LLMs can reason over text, but they need to be told what the psychometric model found.

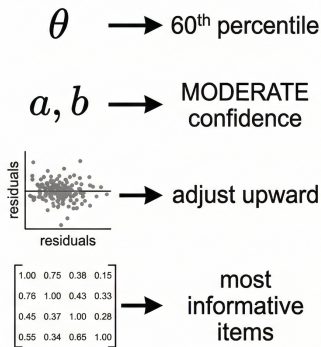
What if we translated psychometric model conclusions into natural language prompts?

The L2L Framework

Psychometric Models



Latent-to-Language Translation



LLM Contextual Reasoning

Psychometrician Persona

Survey psychometrician whose job is to refine model predictions using respondent context

Few-Shot Calibration

3 training examples:
KEEP, ADJUST UP, ADJUST DOWN

Structured Prompt

Respondent at 60th percentile.
Model predicts 4, MODERATE confidence. Correlated items suggest adjusting UPWARD.

Prediction

row_id 42: armed_forces=5,
legislature=4

Experimental Conditions

Data

LAPOP Mexico 2023

12-item trust scale, 7-point | N = 1,430

Latinobarometro Colombia 2023

6-item distrust scale, 4-point | N = 1,073

5-fold cross-validation | 1-4 items masked
per respondent (MCAR)

Imputation Models

LLMs

Claude
Sonnet 4.6

Llama 4
Maverick

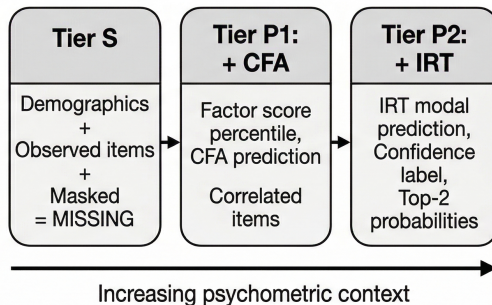
Traditional

mixgb

miceRanger

$m = 5$ imputations, majority vote

Information Tiers



RQ1: Item-Level Measurement Error

Does richer psychometric context reduce prediction error?

$$W_{\kappa} = 1 - \frac{\sum w_{ij} O_{ij}}{\sum w_{ij} E_{ij}}$$

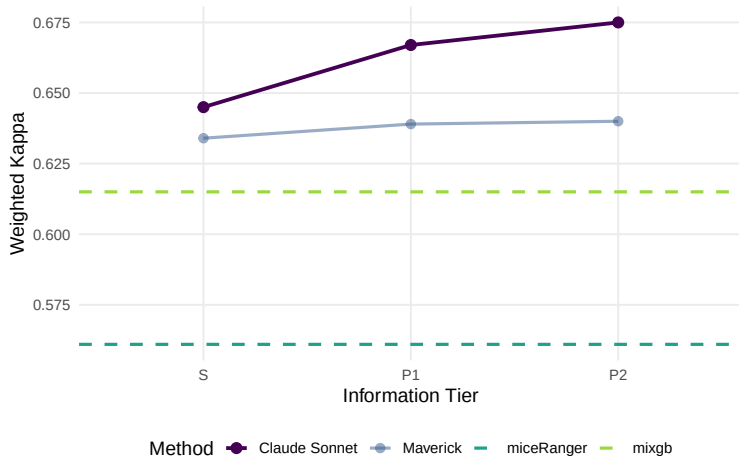
Quadratic weights $w_{ij} = \frac{(i-j)^2}{(K-1)^2}$

$W_{\kappa} = 1$: perfect agreement

Claude S $\rightarrow$ P1 $\rightarrow$ P2:

.645 $\rightarrow$.667 $\rightarrow$.675

Claude outperforms mixgb (.615) and miceRanger (.561) at tier S



RQ2: Person-Level Measurement Error

Do imputed data preserve respondents' latent scores?

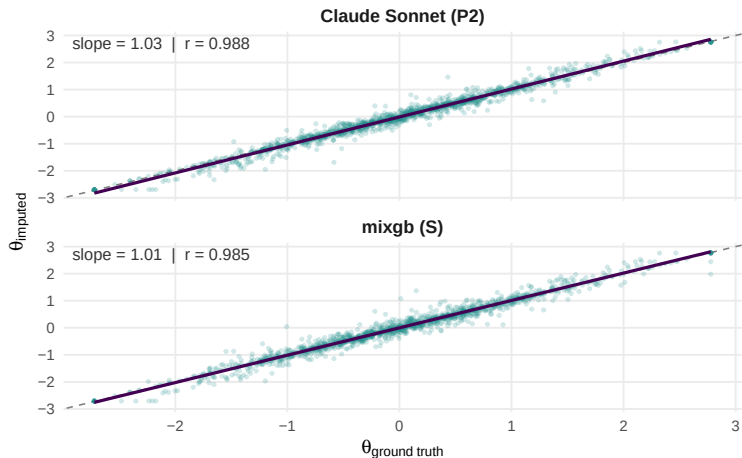
$$\theta_{\text{imp}} = \beta_0 + \beta_1 \theta_{\text{gold}} + \varepsilon$$

β_1 : scale fidelity

$\beta_1 = 1.0$: preserved | < 1.0 : compression

Claude P2 slope = 1.03

mixgb S slope = 1.01



RQ3: Instrument-Level Measurement Error

Does imputation distort the measurement model's psychometric properties?

$$\text{MSE}_j = \overline{\Delta a_j}^2 + \overline{(\Delta a_j - \overline{\Delta a_j})^2}$$

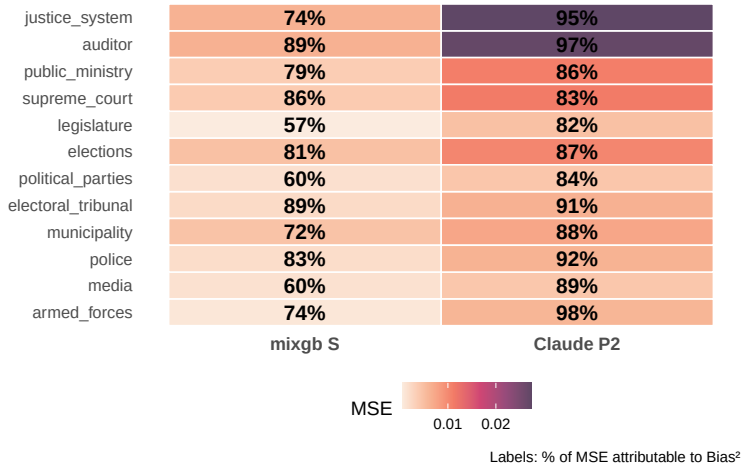
$$\Delta a_{jk} = a_{jk}^{\text{imp}} - a_{jk}^{\text{gold}}$$

High Bias²/MSE: systematic

Low ratio: random noise

Claude P2: 89% systematic
mixgb S: 75%

Both inflate discrimination, but the error breaks down differently



Recommendations and Future Directions

For Survey Methodologists

1. Check for LLM world-knowledge bias at the item level before deploying
2. Less is more: softer psychometric anchors can outperform detailed ones
3. LLM error is predictable; tree-based error is mixed. Choose based on your analysis goals.

Future Directions

1. Combine LLM and tree-based imputations through model stacking
2. Test how prompt specificity interacts with scale length and newer LLMs
3. Extend to realistic missing data patterns and other constructs

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